

Smart University Management Platform

n8n Workflow Automation Documentation

Domain: Education | Platform: n8n Workflow Automation

5 Interconnected Workflows | 28+ Nodes | AI-Powered Decisioning

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Prepared as part of Assignment 3 — Smart University Management Platform

1. Problem Analysis

1.1 Business Context

Universities today manage thousands of students across multiple academic programs, and much of the day-to-day administrative work — registering new students, tracking attendance, collecting and grading assignments, publishing results, and issuing certificates — is still handled manually or through disconnected spreadsheets and email threads. This creates delays, inconsistent record-keeping, and pulls faculty time away from teaching and mentoring. The Smart University Management Platform addresses this by connecting every stage of the student lifecycle — from enrollment to graduation — into a single automated pipeline built on n8n, using tools the university already has access to: Google Forms, Sheets, Drive, Gmail, and AI (Gemini).

1.2 Stakeholders

- **Students** — interact with the system indirectly through form submissions, assignment uploads, and receive automated emails (welcome messages, attendance alerts, grades, certificates).
- **Faculty / Professors** — benefit from automated grading support and are looped in only when a submission needs human judgment (e.g., a low AI-assigned grade).
- **Academic Counselors** — automatically copied on low-attendance alerts so at-risk students can be supported early.
- **University Administrators / Deans** — trigger end-of-term processes (like certificate generation) and receive audit trail visibility into system-wide events and errors.
- **IT / Systems Team** — rely on the centralized audit log and exception alerts to monitor system health without manually checking every workflow.

1.3 Pain Points Identified

- Manual student registration is slow and error-prone, with no consistent ID-generation or record-keeping standard.
- Attendance tracking is typically reviewed too late — students often aren't warned about low attendance until it's a serious problem.
- Grading assignments manually is time-consuming for faculty, delaying feedback to students.
- There is no single source of truth for what happened across different administrative processes — errors go unnoticed until they cause bigger problems.
- Certificate generation at term-end is a repetitive, manual task prone to formatting and mailing mistakes.

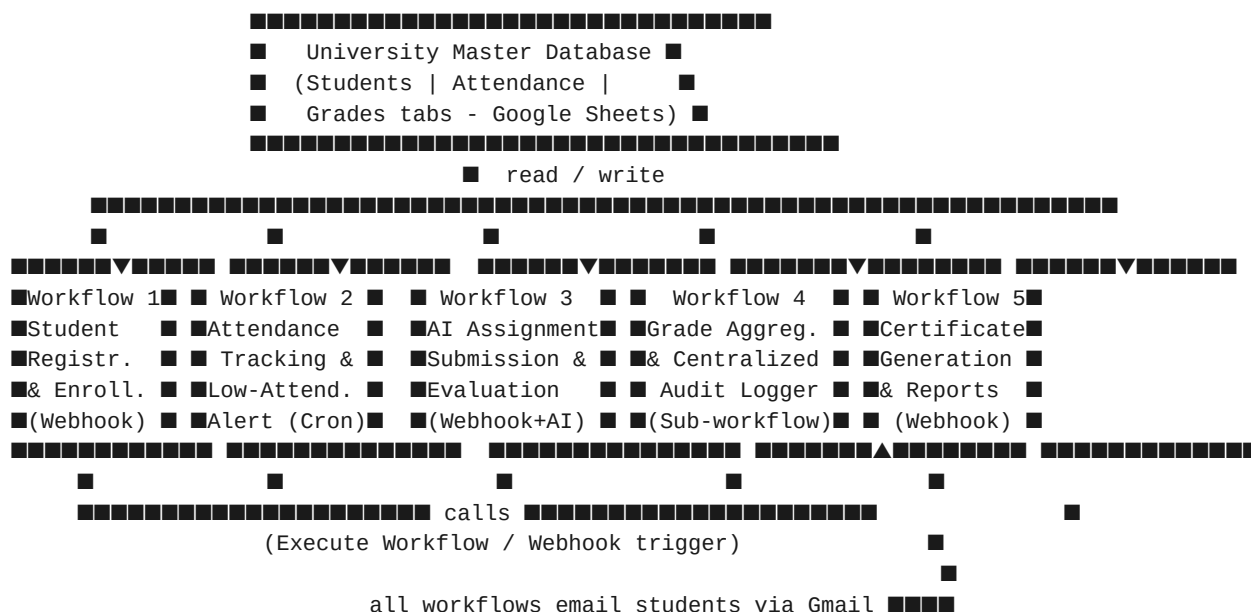
1.4 Objectives

- Digitize and standardize academic administration end-to-end.
- Automate assignment collection, AI-assisted evaluation, and grade publication.
- Proactively detect and alert on low attendance before it becomes a bigger issue.
- Simplify and de-risk end-of-term result and certificate processing.
- Improve communication turnaround between students, faculty, and administrators.
- Maintain a centralized, queryable audit trail of every automated action taken by the platform.

2. Workflow Architecture

2.1 Overall System Architecture

The platform is composed of five independent, purpose-built n8n workflows. Each workflow owns a distinct part of the student lifecycle and can run on its own schedule or trigger, but they are wired together through two shared mechanisms: a common Google Sheets database (the 'University Master Database', with tabs for Students, Attendance, Grades) that every workflow reads from or writes to, and a central Audit Log workflow that other workflows call into whenever a significant event occurs.

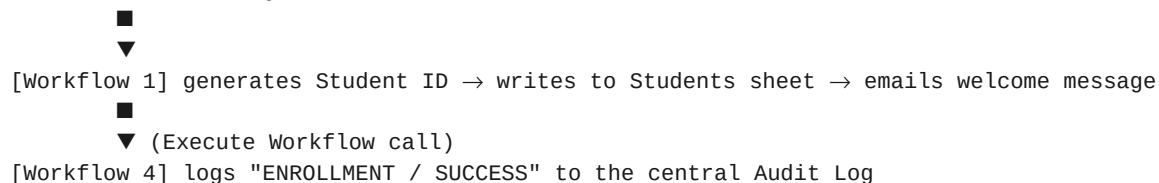


2.2 Workflow Interaction Diagram

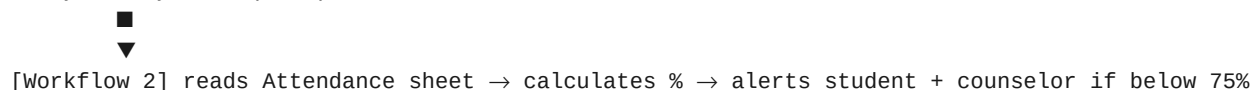
Workflows 1 and 3 actively call into Workflow 4 (the Audit Log) whenever a key event completes — an enrollment or an assignment grade. Workflow 4 is intentionally decoupled: it doesn't know or care which workflow called it, it simply records what happened, timestamps it, classifies its severity, and — for anything marked as an error — immediately alerts an administrator by email. This makes it easy to add more callers in the future (e.g., Workflow 2 or 5 could also log into it) without changing Workflow 4 itself.

2.3 Event Flow Between Workflows

Student submits Google Form



Every Friday 5 PM (Cron)



Student uploads assignment (Webhook)



[Workflow 3] downloads file → extracts text → Gemini AI grades it → parses JSON



■■■ grade < 50 → flagged for human review, still logged & emailed

■■■ grade ≥ 50 → published directly, grade + feedback emailed to student



▼ (Execute Workflow call)

[Workflow 4] logs "ASSIGNMENT_GRADED" to the central Audit Log



Dean triggers term completion (Webhook, manual)



[Workflow 5] fetches eligible students (GPA ≥ 2.0) → loops through each

→ generates certificate PDF → stores in Drive → emails student

3. Workflow-by-Workflow Breakdown

Workflow 1 — Student Registration & Enrollment (5 nodes)

Trigger: A Google Form submission (captured via a linked Google Sheet, since Google Forms has no native webhook, n8n polls the response sheet for new rows using a Google Sheets Trigger node).

How it works: When a student submits the registration form, this workflow generates a unique Student ID, writes the student's record into the central Students sheet, emails the student their ID and portal instructions, and finally logs the successful enrollment into the shared Audit Log (Workflow 4).

#	Node Name	n8n Node Type	Purpose
1	Google Form Submitted (Sheet Trigger)	Google Sheets Trigger	Polls the linked Form Responses sheet for new student submissions.
2	Generate Student ID	Code	Creates a unique STU-XXXX ID and normalizes the form's raw field names.
3	Append Student Row	Google Sheets	Writes the new student's record into the Students tab of the master database.
4	Send Welcome Email	Gmail	Emails the student their generated Student ID and portal setup instructions.
5	Trigger Audit Log	Execute Workflow	Calls Workflow 4 with an ENROLLMENT / SUCCESS event for central logging.

Live n8n Canvas — Successful Test Execution

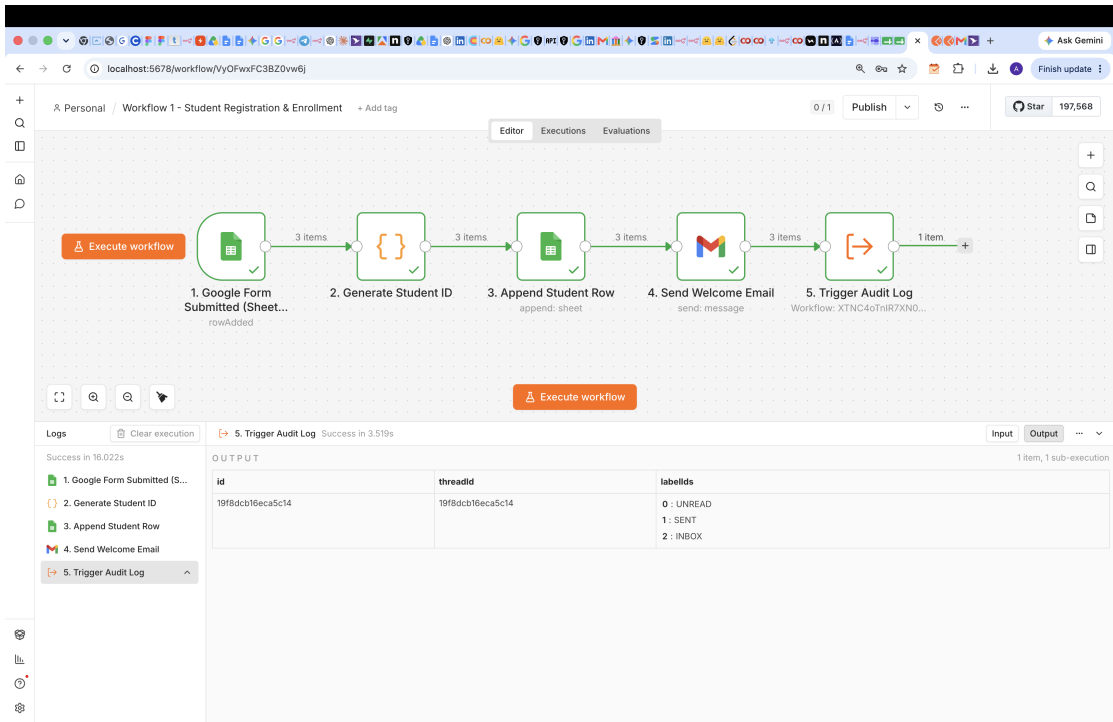


Figure 1 — Workflow 1 executed end-to-end in n8n: form data captured, Student ID generated, row appended, welcome email sent, and the Audit Log successfully triggered.

Workflow 2 — Attendance Tracking & Low-Attendance Alert (5 nodes)

Trigger: A scheduled Cron trigger, running automatically every Friday at 5:00 PM.

How it works: The workflow pulls every student's attendance record, calculates their attendance percentage in code, and — for anyone below the 75% threshold — automatically emails both the student and their academic counselor, so support can be offered before the situation becomes critical.

#	Node Name	n8n Node Type	Purpose
1	Every Friday 5PM	Schedule Trigger (Cron)	Fires automatically once a week to run the attendance check.
2	Read Attendance Sheet	Google Sheets	Fetches every student's Total_Classes and Attended_Classes.
3	Calculate Attendance %	Code	Computes each student's attendance percentage and flags rows below 75%.
4	Is Attendance Low?	IF	Branches the flow — only low-attendance students proceed further.
5	Send Low Attendance Alert	Gmail	Emails the student (CC: counselor) with their current attendance percentage.

Live n8n Canvas — Successful Test Execution

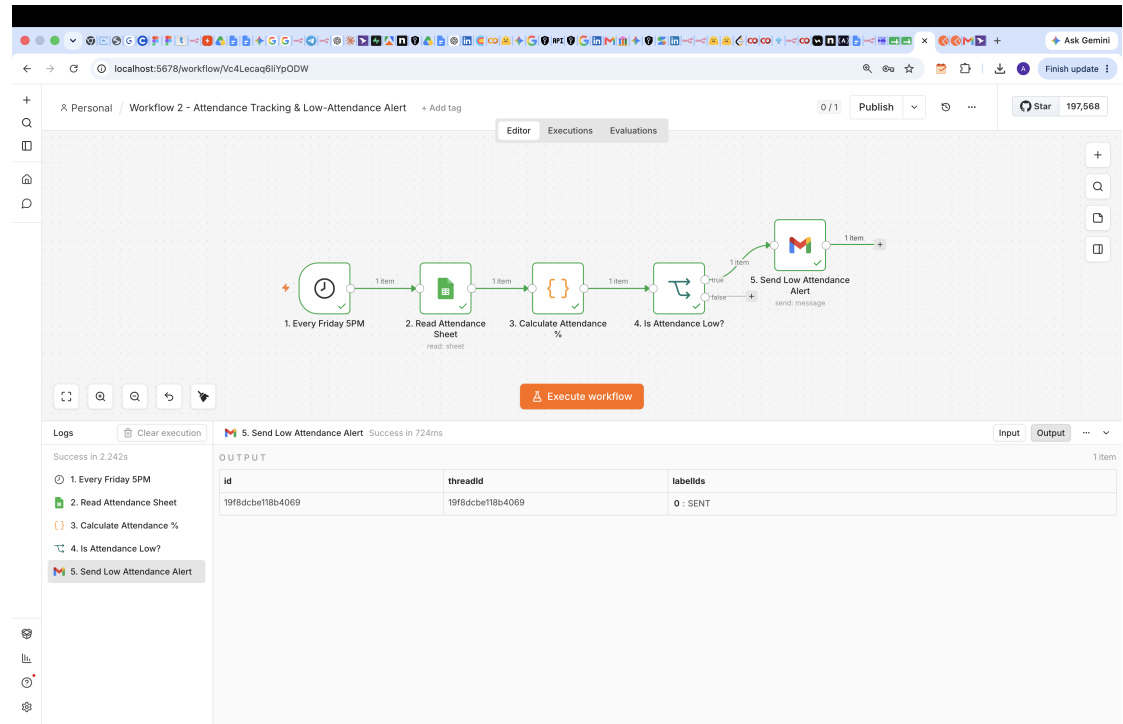


Figure 2 — Workflow 2 executed successfully: attendance read from the sheet, percentage calculated, low-attendance branch triggered, and alert email sent.

Workflow 3 — AI-Powered Assignment Submission & Evaluation (8 nodes)

Trigger: A Webhook fired when a student submits an assignment (with a Google Drive file link, Student ID, and Assignment ID).

How it works: This is the platform's core AI decision-making workflow. It downloads the submitted file from Drive, extracts its plain text, and sends it to Google's Gemini AI with a grading prompt. Gemini returns a numeric grade and constructive feedback as JSON, which is parsed and checked against a threshold: grades below 50 are automatically routed for human faculty review rather than being published outright, while

passing grades are saved and emailed to the student immediately. This is the workflow's human approval gate — the AI can grade autonomously, but low-confidence outcomes are never silently auto-published.

#	Node Name	n8n Node Type	Purpose
1	Assignment Uploaded (Webhook)	Webhook	Captures the student's submission link, Student ID, and Assignment ID.
2	Download Assignment File	Google Drive	Downloads the submitted file's binary content from Drive.
2b	Extract Text From File	Extract From File	Converts the downloaded binary into plain text for the AI to read.
3	AI Auto-Grading & Feedback (Gemini)	HTTP Request	Sends the assignment text to Gemini with a grading prompt; receives a JSON grade + feedback.
4	Parse AI Output	Code	Parses Gemini's JSON response and flags submissions scoring below 50 for human review.
5	Grade Below 50? (Human Review Gate)	IF	Routes low-confidence grades to manual review instead of auto-publishing.
6	Save Grade to Sheet	Google Sheets	Records the Student ID, Assignment ID, grade, feedback, and status into the Grades tab.
7	Publish Grade to Student	Gmail	Emails the student their grade and AI-generated feedback, or a 'pending review' notice.

Live n8n Canvas — Successful Test Execution

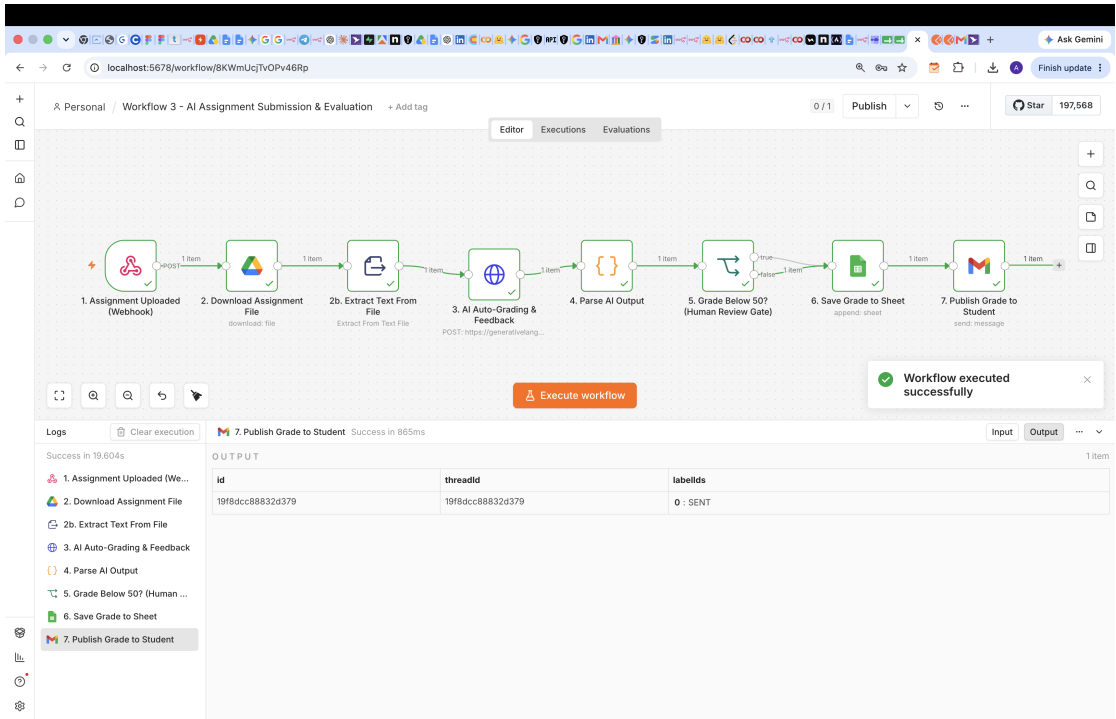


Figure 3 — Workflow 3 executed successfully end-to-end: assignment downloaded, text extracted, Gemini AI grading completed, grade parsed and saved, and the result emailed to the student.

Workflow 4 — Grade Aggregation & Centralized Audit Logger (6 nodes)

Trigger: Called internally by other workflows (via an Execute Workflow Trigger) whenever a significant event needs to be recorded — it never runs on its own schedule.

How it works: This workflow is the platform's central nervous system for observability. Any other workflow can hand it an event, and it standardizes the incoming fields, formats a structured log entry, writes it to a single central Audit Log sheet, and then routes the event by severity: routine Info and Warning-level events are simply recorded, while Critical/Error-level events additionally trigger an immediate escalation email to the admin team, so failures are never silently missed.

#	Node Name	n8n Node Type	Purpose
1	Workflow Trigger	Execute Workflow Trigger	Receives an event payload (action, status, details, severity) from another workflow.
2	Edit Fields	Set	Normalizes and structures the incoming event fields before formatting.
3	Format Log Entry	Code	Builds a standardized log entry, attaching a timestamp and consistent formatting.
4	Append to Audit Log	Google Sheets	Appends the formatted event into a shared Audit Log tab for full traceability.
5	Severity Filter	Switch	Routes the event down one of three paths — INFO, WARNING, or CRITICAL_ERROR.
6	Escalation Alert (Gmail)	Gmail	Fires only on the WARNING/CRITICAL_ERROR path, immediately notifying admin staff by email.

Live n8n Canvas — Successful Test Execution

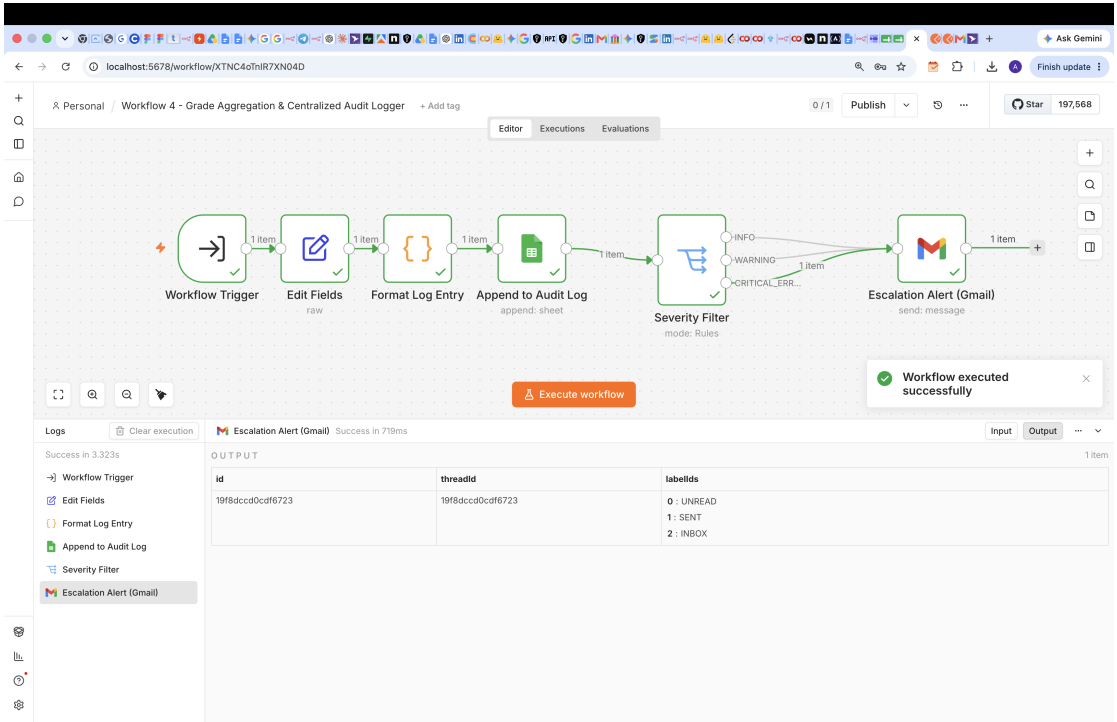


Figure 4 — Workflow 4 executed successfully: incoming event normalized, logged to the central Audit Log sheet, routed by severity, and an escalation email sent for the CRITICAL_ERROR path.

Workflow 5 — End-of-Term Certificate Generation & Reports (6 nodes)

Trigger: A Webhook, fired manually by a Dean or Administrator specifying a Program Code and Graduation Year.

How it works: This workflow automates what is traditionally one of the most repetitive end-of-term tasks. It queries the master database for every student who is eligible to graduate (active status and GPA at or above 2.0), then loops through them one at a time — generating a personalized certificate as HTML, converting it into a real PDF, archiving it to a Google Drive folder, and emailing it directly to the student — before moving on to the next eligible student.

#	Node Name	n8n Node Type	Purpose
1	Term Completion Triggered (Webhook)	Webhook	Manually triggered by an admin with a Program Code and Graduation Year.
2	Fetch Eligible Students	Google Sheets	Filters the Students tab for matching Program, Graduation Year, and Eligible = TRUE.
3	Loop Over Eligible Students	Loop Over Items (Split In Batches)	Iterates through each eligible student one at a time.
4	Generate Certificate PDF	Code + HTTP Request	Builds a personalized certificate as HTML and converts it into a real PDF file.
5	Store Certificate in Drive	Google Drive	Uploads the generated PDF into the university's Certificates folder.
6	Email Certificate to Student	Gmail	Sends the student their official Certificate of Completion as an attachment.

Live n8n Canvas — Successful Test Execution

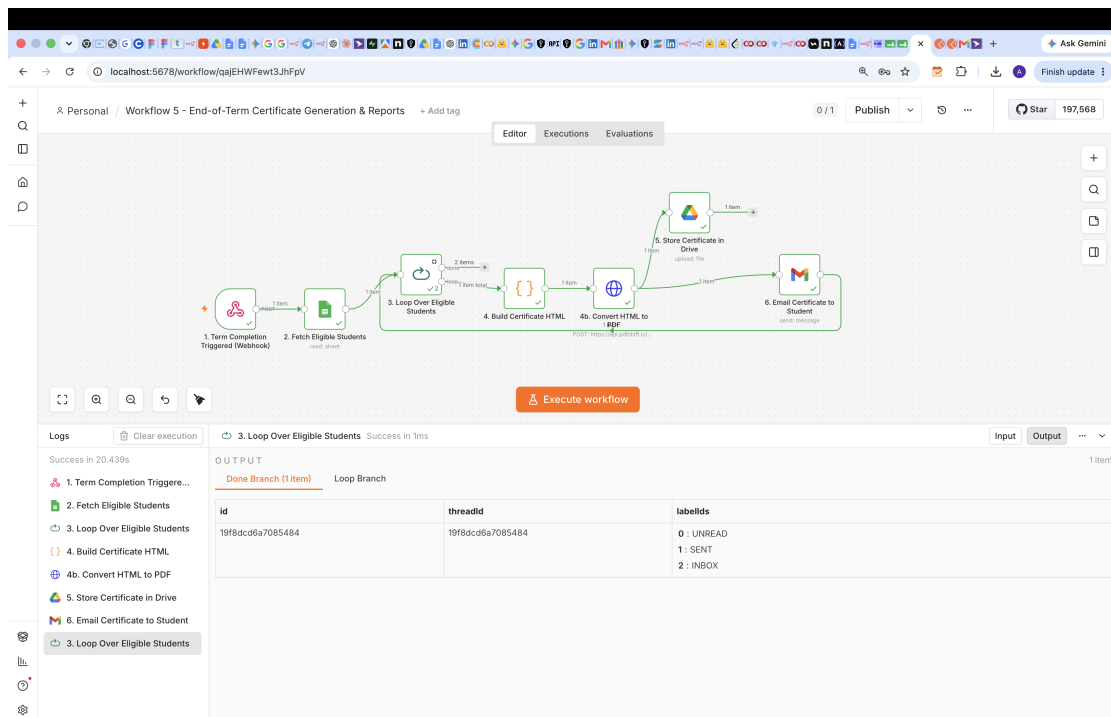


Figure 5 — Workflow 5 executed successfully: eligible students fetched, looped through, certificate PDF generated and stored in Drive, and emailed to the student.

4. Advanced Features Implemented

Feature	How It Is Implemented
AI-Powered Decision Making	Workflow 3 uses Google Gemini to autonomously grade assignments and generate feedback, removing manual grading effort for straightforward submissions.
Human Approval Gate	Workflow 3's IF node routes any AI grade below 50 to manual faculty review instead of auto-publishing, keeping a human in the loop for low-confidence outcomes.
Error Handling & Central Logging	Workflow 4 acts as a dedicated audit and exception logger, called by other workflows to record every significant event and immediately alert admins on failures.
Logging & Audit Trail	Every enrollment and grading event is timestamped and written to a single central Audit Log sheet, giving administrators full traceability across the platform.
Scheduled Workflows (Cron)	Workflow 2 runs automatically every Friday at 5 PM with no manual intervention required.
Webhook-Triggered Workflows	Workflows 1, 3, and 5 are all event-driven via webhooks — form submissions, assignment uploads, and manual admin actions respectively.
Conditional Branching	IF and Switch nodes are used throughout (attendance thresholds, grade thresholds, log severity) to route data down different paths automatically.
Loops / Batch Processing	Workflow 5 uses a Loop Over Items node to process every eligible graduating student sequentially, generating and sending one certificate at a time.

5. Summary

The Smart University Management Platform brings together five independent but interconnected n8n workflows that automate the full academic administration lifecycle — from the moment a student registers, through attendance monitoring, assignment grading, and finally graduation and certification. Rather than being five disconnected scripts, the workflows share a common data backbone (the Google Sheets master database) and a common accountability layer (the central Audit Log in Workflow 4), so the system behaves as a single cohesive platform rather than a collection of unrelated automations.

In total, the platform spans 28+ individual nodes across the five workflows, and demonstrates every major automation pattern expected of a production-grade system: scheduled and webhook triggers, AI-assisted decision-making with a human fallback, conditional branching, batch looping, and centralized error handling and logging.

How This Helps in Daily University Life

- **For students:** Registration is instant and paperless, attendance issues are flagged early enough to actually do something about them, assignment feedback arrives within minutes instead of days or weeks, and certificates arrive automatically at the end of the program without needing to visit an admin office.
- **For faculty:** Routine, low-stakes grading is handled automatically, freeing up time for teaching and for the assignments that genuinely need a human's judgment (the ones the system flags for review).
- **For academic counselors:** Automatic attendance alerts mean at-risk students can be reached out to proactively, rather than counselors having to manually audit spreadsheets every week.
- **For administrators and the Dean's office:** End-of-term certificate generation — traditionally a manual, error-prone bottleneck — becomes a single button-press that handles an entire graduating batch unattended.
- **For IT / operations:** The centralized audit trail means nobody has to guess what happened when something goes wrong — every workflow's key actions are timestamped and searchable in one place, with real-time alerts the moment something fails.

Overall, the platform doesn't just digitize existing paperwork — it removes entire categories of manual, repetitive work from people's daily routines, while still keeping a human firmly in control wherever real judgment is required (grade review, term-completion sign-off, and error escalation).